

Multi-Agent Protocol Implementation Plan

Capstone Project: Agentic AI Protocols (MCP, A2A, ACP)

To demonstrate comprehensive mastery of multi-agent communication protocols, you will create a detailed implementation plan for a real-world multi-agent system that strategically leverages MCP, A2A, and ACP protocols.

Project Overview

You will analyze a complex organizational scenario, identify communication and coordination challenges, and design a comprehensive multi-protocol solution that addresses these challenges effectively. Your implementation plan will serve as a blueprint that could be used by development teams to build the system you've designed.

Scenario

Select one of the following complex organizational scenarios:



1. Smart City Emergency Response System: Coordinate police, fire, medical, traffic management, and public communication agents during emergency situations.

2. Global Supply Chain Optimization Platform: Manage supplier agents, logistics coordinators, inventory managers, demand forecasters, and quality control agents across multiple countries.

3. Healthcare Patient Care Coordination System: Orchestrate primary care physicians, specialists, diagnostic labs, pharmacy, billing, and insurance agents for complex patient care workflows.

4. Enterprise Software Development Platform: Coordinate project management, code analysis, testing, deployment, and monitoring agents for continuous integration and delivery.

5. Financial Trading and Risk Management System: Manage market analysis, trading execution, risk assessment, compliance monitoring, and reporting agents for institutional trading.

Instructions for learners

Implementation Plan Structure

Part 1: Scenario Analysis and Requirements (25%)

- Comprehensive analysis of your chosen scenario
- Identification of key stakeholders and their communication needs
- Definition of performance, security, and scalability requirements
- Analysis of current communication challenges and their business impact

Part 2: Protocol Selection and Architecture Design (35%)

- Strategic selection of MCP, A2A, and ACP protocols for different system components
- Detailed justification for each protocol choice based on specific requirements
- Integration architecture showing how protocols work together
- Communication flow diagrams for critical workflows
- Identification of protocol handoff points and transition mechanisms

Part 3: Implementation Strategy and Optimization (25%)

- Phased implementation plan with clear milestones and deliverables
- Performance optimization strategies for each protocol
- Security and compliance considerations specific to your scenario
- Scalability planning for system growth and evolution
- Risk mitigation strategies for potential implementation challenges

Part 4: Monitoring and Evaluation Framework (15%)

- Key performance indicators for measuring system effectiveness
- Monitoring strategies for each protocol's performance
- Evaluation methods for overall system success
- Continuous improvement processes for protocol optimization

Deliverables

Submit a comprehensive implementation plan (3,000-4,000 words) that includes:

- Executive summary with key recommendations
- Detailed technical architecture with diagrams
- Risk analysis and mitigation strategies
- Success metrics and evaluation criteria

Evaluation Criteria

- Technical Depth: Demonstrates comprehensive understanding of MCP, A2A, and ACP protocols
- Strategic Thinking: Shows appropriate protocol selection based on specific requirements
- Integration Complexity: Effectively designs multi-protocol architectures
- Practical Viability: Proposes realistic implementation strategies
- Professional Quality: Presents findings in a format suitable for executive decision-making

Completion

This capstone project represents the culmination of your learning journey, demonstrating your ability to analyze complex scenarios, make strategic protocol decisions, and design comprehensive multi-agent communication systems that solve real-world problems.